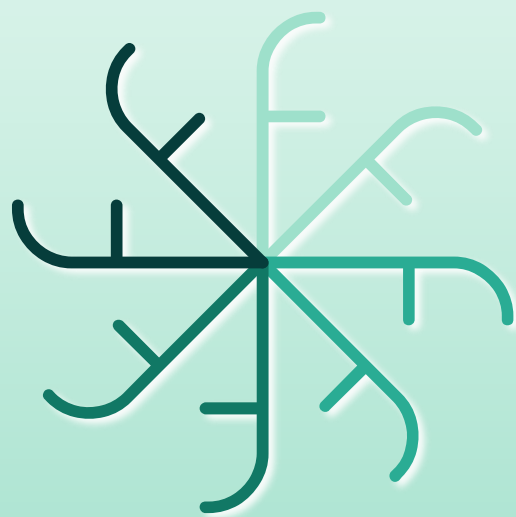




fabric
FEBRUARY



Extending Fabric monitoring for custom requirements

Iqbal Khan & Marc Lelijveld

Presenters

Marc Lelijveld

Technical Evangelist & Data Platform MVP

- Joined Macaw in 2018
- Accidental Fabric Administrator at various enterprise clients
- Previously only focused on Power BI, till Fabric came along!
- Based in a tiny town next to Gouda – Netherlands



Presenters

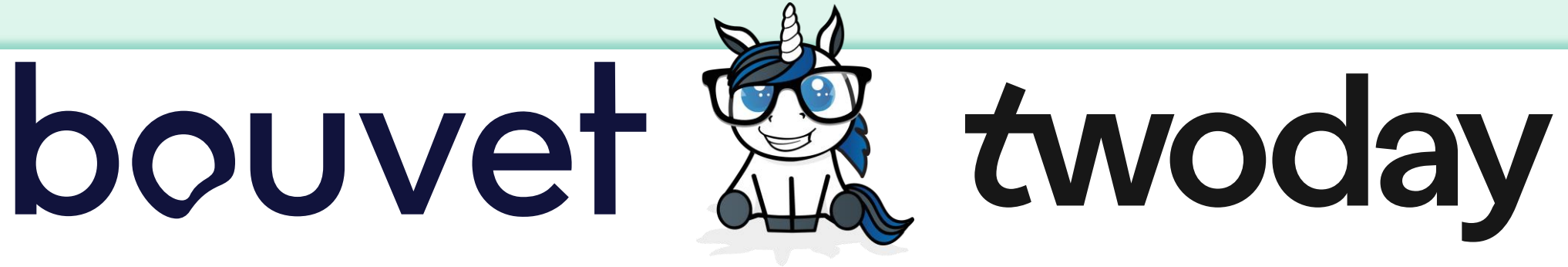
Iqbal Khan

Admin Monitoring + OneLake Catalog —
Reporting Lead @ Microsoft

- Joined Microsoft in 2021 as a Power BI architect
- Moved to Fabric engineering in 2023
- Based in NYC & becoming an uncle today!



Thank you to our Fabric February Friends!



Agenda – Extending Fabric monitoring for custom requirements

- 1 Why monitoring? Which monitoring solution is right for me?
- 2 OneLake Catalog + Admin monitoring workspace deep-dive
- 3 Custom monitoring through Semantic Link
- 4 Session closeout, Q/A



Why monitoring? Monitoring can help with:



Best practice and
policy enforcement



Audit and compliance



Efficiency and
reusability



Cost optimization



Why monitoring?

As organizations continue to expand their use of Fabric, it is critical to have a governance strategy in place to ensure secure and effective use of your data.

Monitoring can help with:

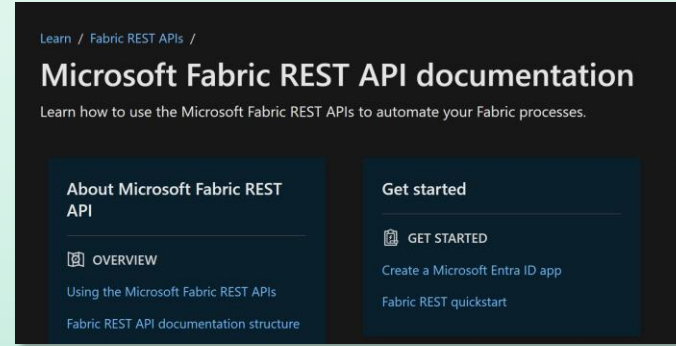
- Understanding data solutions needed to support business
 - Ensuring safe and secure use of data
 - Maximizing Fabric return on investment
 - Reducing data outages
 - Enforcing best practices and organizational policies
- ...and much more!



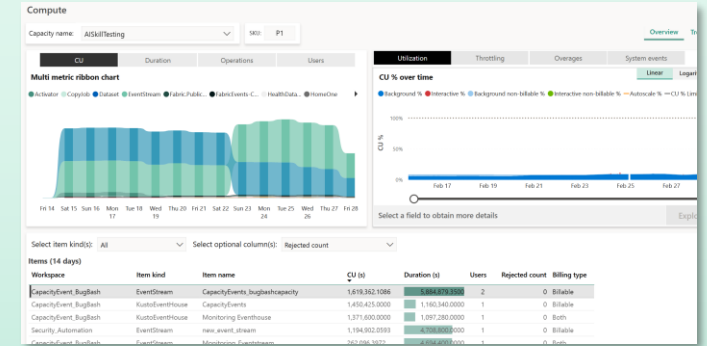
Which Microsoft monitoring solution is right for me?



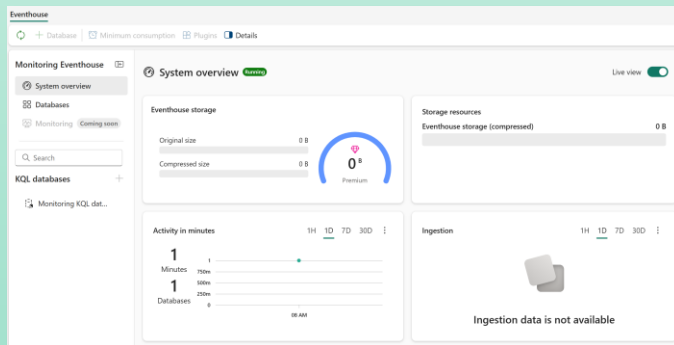
Admin monitoring – Out of the box reports on key governance scenarios | **Admin**



APIs – Flexible, programmatic data extraction & automation | **Admin\Developer**



Capacity Metrics App – Capacity management and resource planning | **Tenant\Capacity Admin**



Workspace Monitoring – Diagnostic logging for Fabric items | **Developer**

#FabricFebruary

Real-Time hub

All data streams

This lists the data streams you can access. Click a stream to see more details.

Search for streaming data across your organization

Name	Owner	Item	Workspace
capacity_utilization_event_stream-stream	Fabric Admin	capacity_utilization_event_st	CapacityEvent_Bugbash
new_event_stream-stream	Fabric Admin	new_event_stream	Security_Automation
Weather	Fabric Admin	OsherEventhouse	osher222
Monitoring_Eventstream-stream	—	Monitoring_Eventstream	CapacityEvents_Demo
Monitoring_Eventstream-stream	—	Monitoring_Eventstream	CapacityEvent_Bugbash
CapacityEventAll	Fabric Admin	CapacityEvents	CapacityEvent_Bugbash
CapacityEvents_bugbashcapacity-stream	Fabric Admin	CapacityEvents_bugbashcap	CapacityEvent_Bugbash
azure_storage_event_stream-6-stream	Fabric Admin	azure_storage_event_stream	Varun_test

Real Time Hub – Event streaming for Fabric workloads for responsive actions | **Developer**

OneLake catalog

Domain: All domains

Explore Govern (preview)

Name	Type	Owner	Refreshed	Location	Endorsement
OneLake catalog governance repo.	Semantic model	Fabric Admin	2/28/25, 8:27:26 ...	My Workspace	—
Monitoring Eventhouse	Eventhouse	—	—	ADMIN AND GOV P...	—
FeatureUsageAndAdoption 6	Semantic model	Fabric Admin	2/28/25, 7:55:58 ...	Admin Analytics (D...	—
App Id App Name Artifact Uri Add...	Semantic model	Fabric Admin	2/28/25, 7:37:5...	Admin Analytics (D...	—
AmazonPurchases2016	Semantic model	Fabric Admin	1/15/25, 4:46:42 ...	kv12	—
FinacialPBX (1) 2	Semantic model	Fabric Admin	1/2/25, 6:11:03 ...	VZWS-Raki-02-01-25	—
Corporate Spend	Semantic model	Fabric Admin	1/2/25, 2:00:56 ...	VZWS-Raki-02-01-25	—
sample file	Semantic model	Fabric Admin	10/9/24, 4:28:57 ...	ad_09_Oct	—
RetailRader	Semantic model	Fabric Admin	10/16/24, 12:3...	IgniteDemo2024	—

OneLake catalog – Actionable lens into your data estate | **Developer**



OneLake Catalog — For developers

Public preview

Central hub to find, explore, and manage the Fabric items you own

Overview (Explore tab)

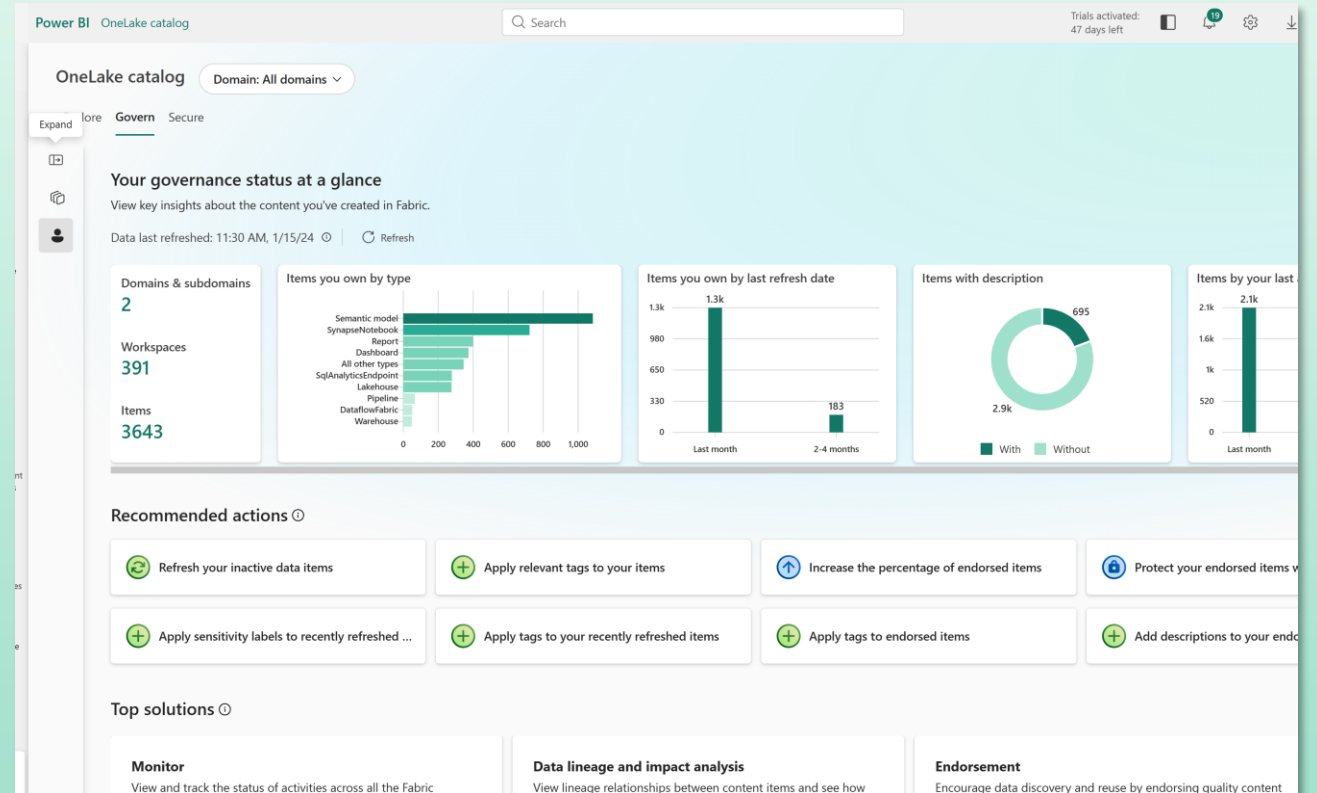
OneDrive-like UI to browse and explore **all your Fabric items** in one place without losing context

Item Details (Explore tab contd.)

Dive further into item details, highlighting last refresh time/status, permissions, external shares, lineage, and more

Act (Govern tab)

Dynamic actions to better secure and govern your data, plus built-in insights to visualize your data estate



OneLake Catalog – For admins

Public preview

Central hub to set up, manage, and protect the Fabric items across your entire tenant

Overview (Explore tab)

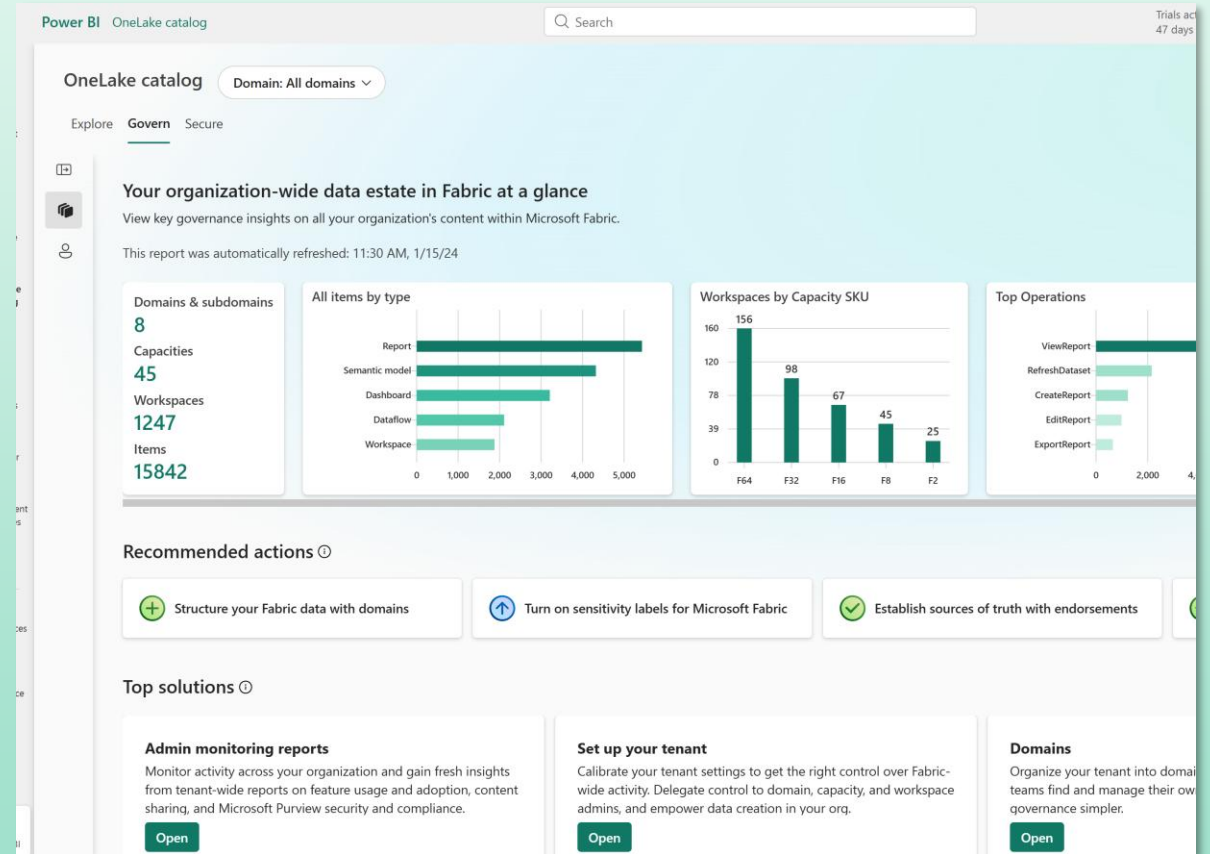
OneDrive-like UI to browse and explore **your entire data estate** in one place without losing context

Item Details (Explore tab contd.)

Dive further into item details, highlighting last refresh time/status, permissions, external shares, lineage, and more

Take action (Govern tab)

Dynamic actions to better secure and govern **your entire data estate**, plus built-in insights to support governance efforts



**Not covered: Secure tab -> Managing permissions

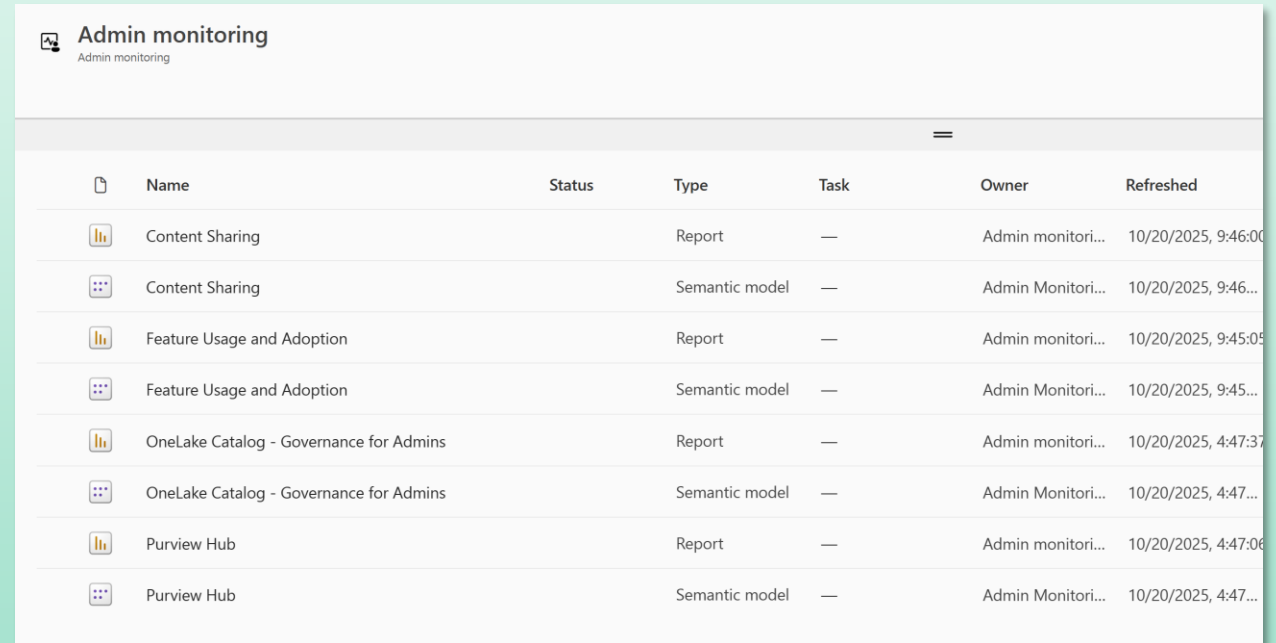
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What is the Admin monitoring workspace?

Public preview

- **One-stop shop** for enterprise reporting and analytics
- Includes **out of the box reporting** focused on Fabric tenant management scenarios
- Also includes **semantic models for customization**
- Available to **all tenants** regardless of licensing type or total # of users



The screenshot displays the 'Admin monitoring' workspace. At the top, there's a header with the title 'Admin monitoring' and a subtitle 'Admin monitoring'. Below the header is a table with the following columns: Name, Status, Type, Task, Owner, and Refreshed. The table contains eight rows of data, each representing a monitoring item. The items are grouped into three categories: Content Sharing, Feature Usage and Adoption, and OneLake Catalog - Governance for Admins. Each category has a 'Report' and a 'Semantic model' entry. The 'Purview Hub' category also has a 'Report' and a 'Semantic model' entry. The 'Status' column for all items is empty. The 'Task' column for all items contains a dash. The 'Owner' column for all items is 'Admin monitoring'. The 'Refreshed' column shows the last refresh time for each item, all of which are from 10/20/2025.

	Name	Status	Type	Task	Owner	Refreshed
	Content Sharing		Report	—	Admin monitoring	10/20/2025, 9:46:00
	Content Sharing		Semantic model	—	Admin Monitoring	10/20/2025, 9:46:00
	Feature Usage and Adoption		Report	—	Admin monitoring	10/20/2025, 9:45:05
	Feature Usage and Adoption		Semantic model	—	Admin Monitoring	10/20/2025, 9:45:05
	OneLake Catalog - Governance for Admins		Report	—	Admin monitoring	10/20/2025, 4:47:37
	OneLake Catalog - Governance for Admins		Semantic model	—	Admin Monitoring	10/20/2025, 4:47:37
	Purview Hub		Report	—	Admin monitoring	10/20/2025, 4:47:06
	Purview Hub		Semantic model	—	Admin Monitoring	10/20/2025, 4:47:06



Perfect – yet imperfect...

There are always custom requirements!



What Microsoft provides

Platform structure and governance, like capacities, domains and workspaces

Item metadata, describing platform artifacts

Audit logs, showing facts about usage and adoption of the platform



What needs customization

Organizational structure

Licensing (part of M365 admin)

Cost monitoring components (capacity costs)

Extensions to monitor cross-platform adoption



Enriching with your own data

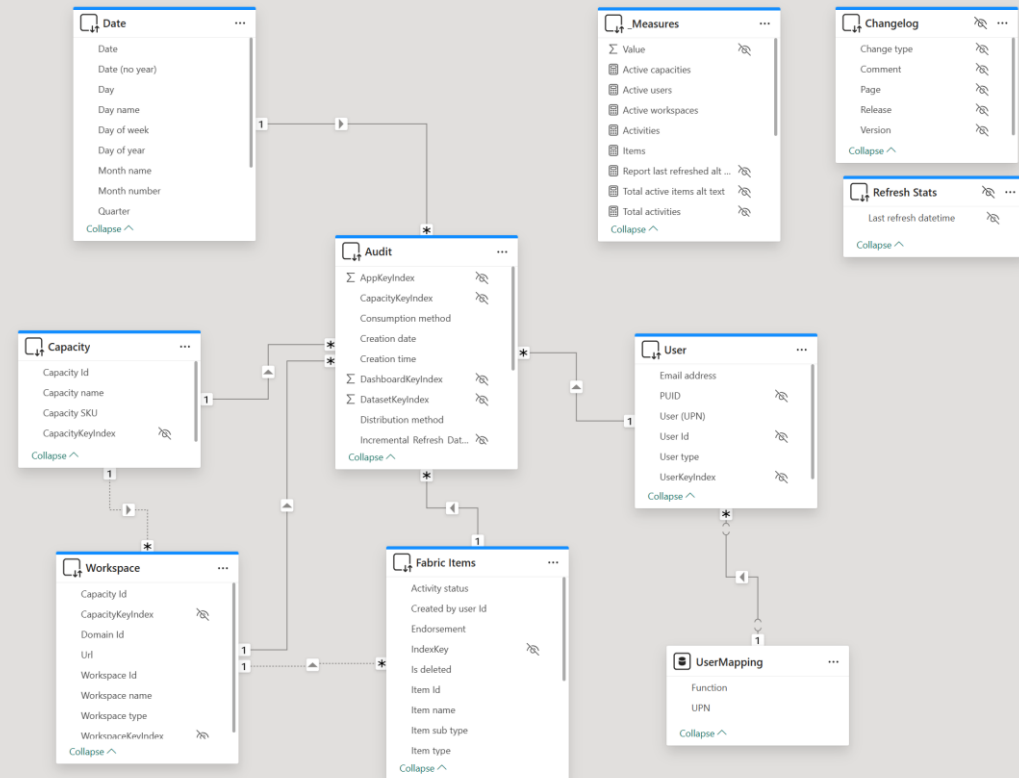
Composite models

What are composite models:

Connect to Power BI semantic models and Azure Analysis Services models using DirectQuery to build custom models using multiple sources

In composite models, you combine 2 or more **source groups** into a single semantic model

Notice two different source groups → Admin monitoring data + custom org hierarchy



Extending Admin monitoring



Goals

- Easily combine data like org. hierarchy, licenses, etc.
- Trend analysis
- Long term storage of data



Non-goals

- Exporting to unmanaged files and carriers (like Excel)
- Distributing data across the organization

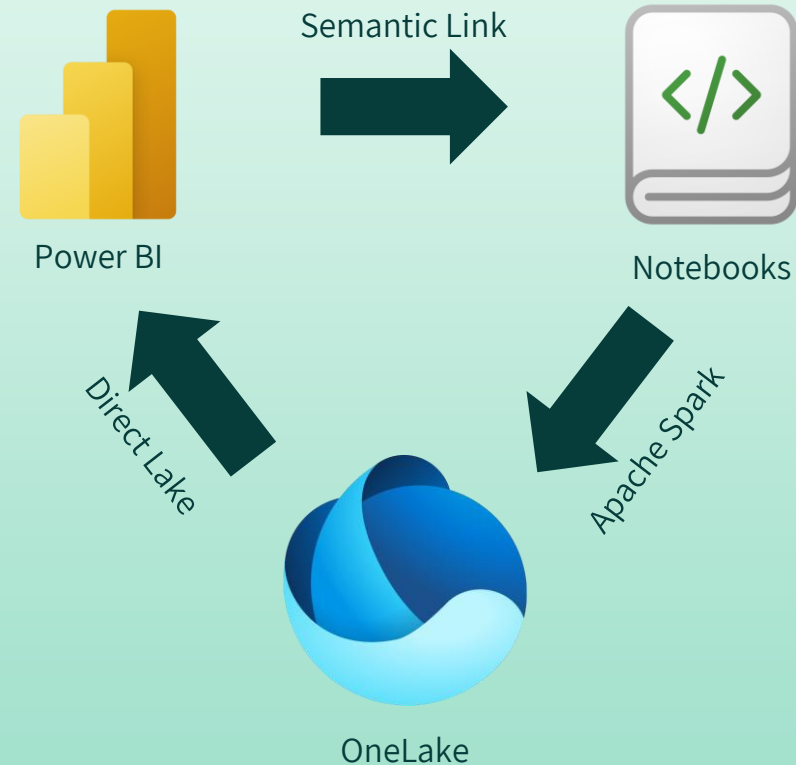


Introduction to Semantic Link

Semantic Link is a Microsoft Fabric feature that allows you to connect to Power BI Semantic Models from Fabric Notebooks.

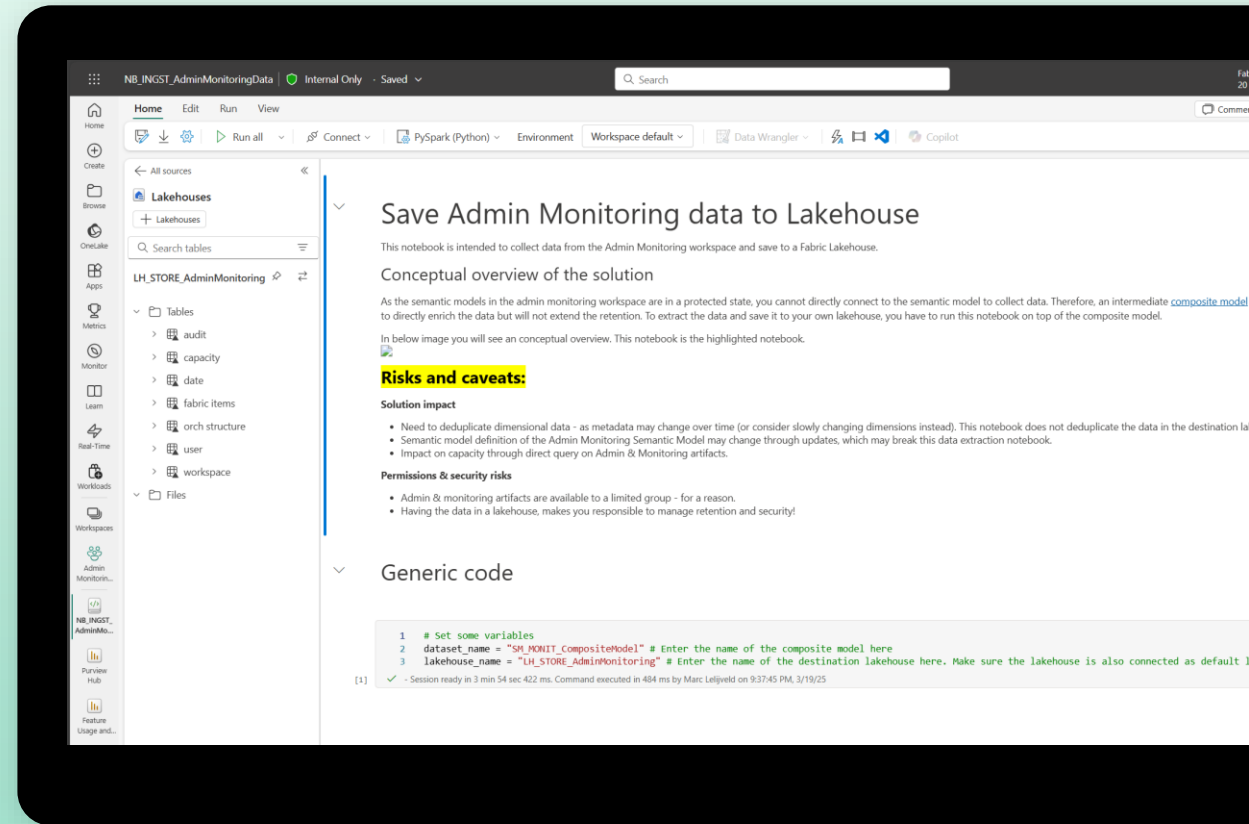
This feature **only** exists and works in Fabric.

- Write DAX, Python, SQL, and informative queries directly against your semantic models
- Data as well as metadata
- Operate and automate tasks like refreshes, partitioning, creation of items, etc.
- More than just semantic models, enables automation of Fabric tasks in general
- Open-source extension Semantic Link Labs to do even more!

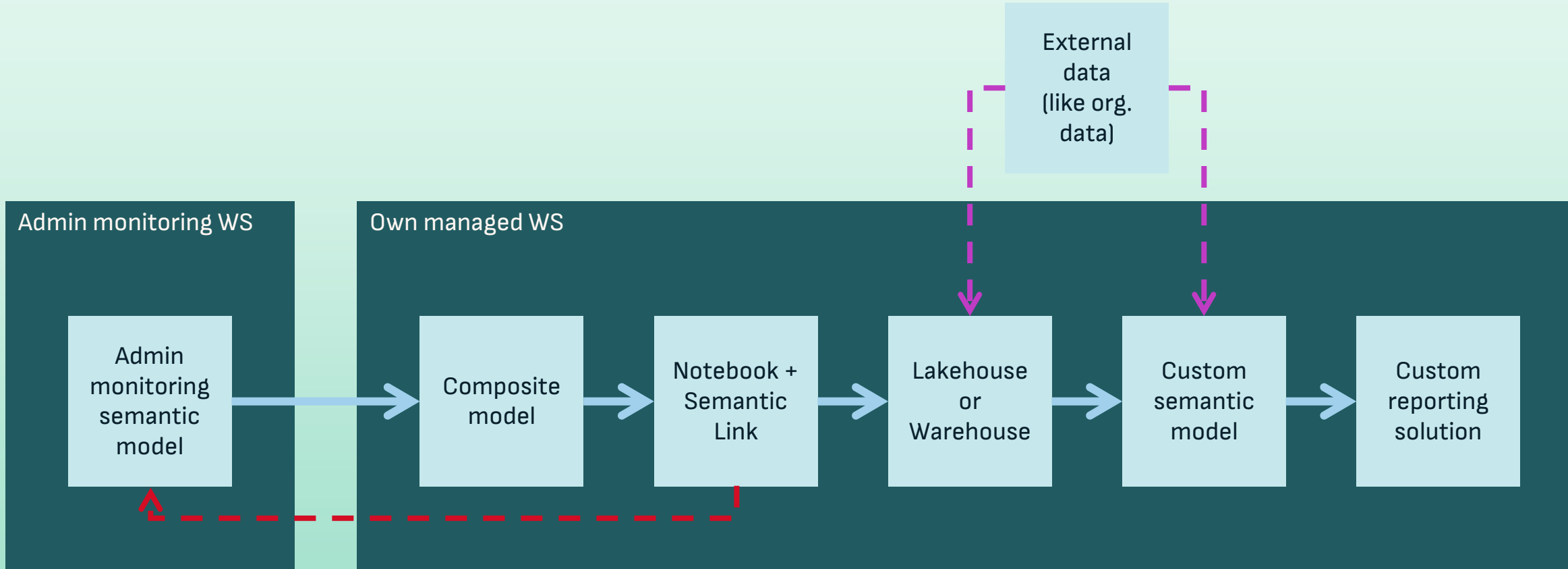


Quick look at Notebooks

- Code first
- Web-based interface
- Cell-based code blocks
- Runs on nodes (part of Fabric capacity)
- Often used languages incl. Python, Spark & Markdown
- Used by data engineers for data ingestion, prep, and transformations
- Used by data scientists for experiments and models



Solution overview (conceptual)



Not permitted

Risks and caveats

Solution impact

- Need to **deduplicate** dimensional data – as metadata may change over time (or consider slowly changing dimensions instead)
- Semantic model definition may **change** through updates, which may **break** your data extraction notebook
- Impact on **capacity** through DirectQuery on Admin monitoring semantic models

Permissions & security risks

- Admin monitoring artifacts are available to a **limited group** – for a reason
- Having the data in a lakehouse makes you responsible to manage **retention** and **security**!



Sharing your admin insights?

Sharing is caring!

Every domain / workspace admin (should) want(s) insights on their own content

Only share what is relevant to them, with customized security on top (RLS / OLS)

But...

After every update, your security config might be overwritten

Lot of manual effort to reconfigure every time

Link: <https://data-marc.com/2025/10/14/automating-power-bi-semantic-model-security-with-fabric-semantic-link/>



**Semantic Link
to the rescue!**

Additional links and resources

Official public documentation:

- Admin Monitoring Workspace
<https://learn.microsoft.com/en-us/fabric/admin/monitoring-workspace>
- Feature Usage and Adoption Report
<https://learn.microsoft.com/en-us/fabric/admin/feature-usage-adoption>
- OneLake Catalog – Overview
<https://learn.microsoft.com/en-us/fabric/governance/onelake-catalog-overview>
- OneLake Catalog – Govern your data
<https://learn.microsoft.com/en-us/fabric/governance/onelake-catalog-govern>



Additional links and resources

Open-source solutions and accelerators:

- Marc's GitHub with scripts for extending Fabric monitoring

<https://github.com/marclelijveld/Fabric-Automation/>

- Fabric Toolbox

<https://github.com/microsoft/fabric-toolbox/tree/main>

- Fabric Unified Admin Monitoring (FUAM)

<https://github.com/microsoft/fabric-toolbox/tree/main/monitoring/fabric-unified-admin-monitoring>

- Fabric Cost Analysis

<https://github.com/microsoft/fabric-toolbox/blob/main/monitoring/fabric-cost-analysis>



Share **your thoughts** and help our speakers!



fabfeb.app/feedback



Thank you!

